

CRC5 FSM

Handshake signals:

bit_{out} → bit_{in} input for BS FSM

done → done input for NRZI FSM

send0 ← output from BS

sendADDR → send0addr input for BS

sendENDP → send0endp input for BS

sendCRC → send0crc input for BS

* Note that the CRC is computed combinationally each cycle of ADDR and ENDP where:

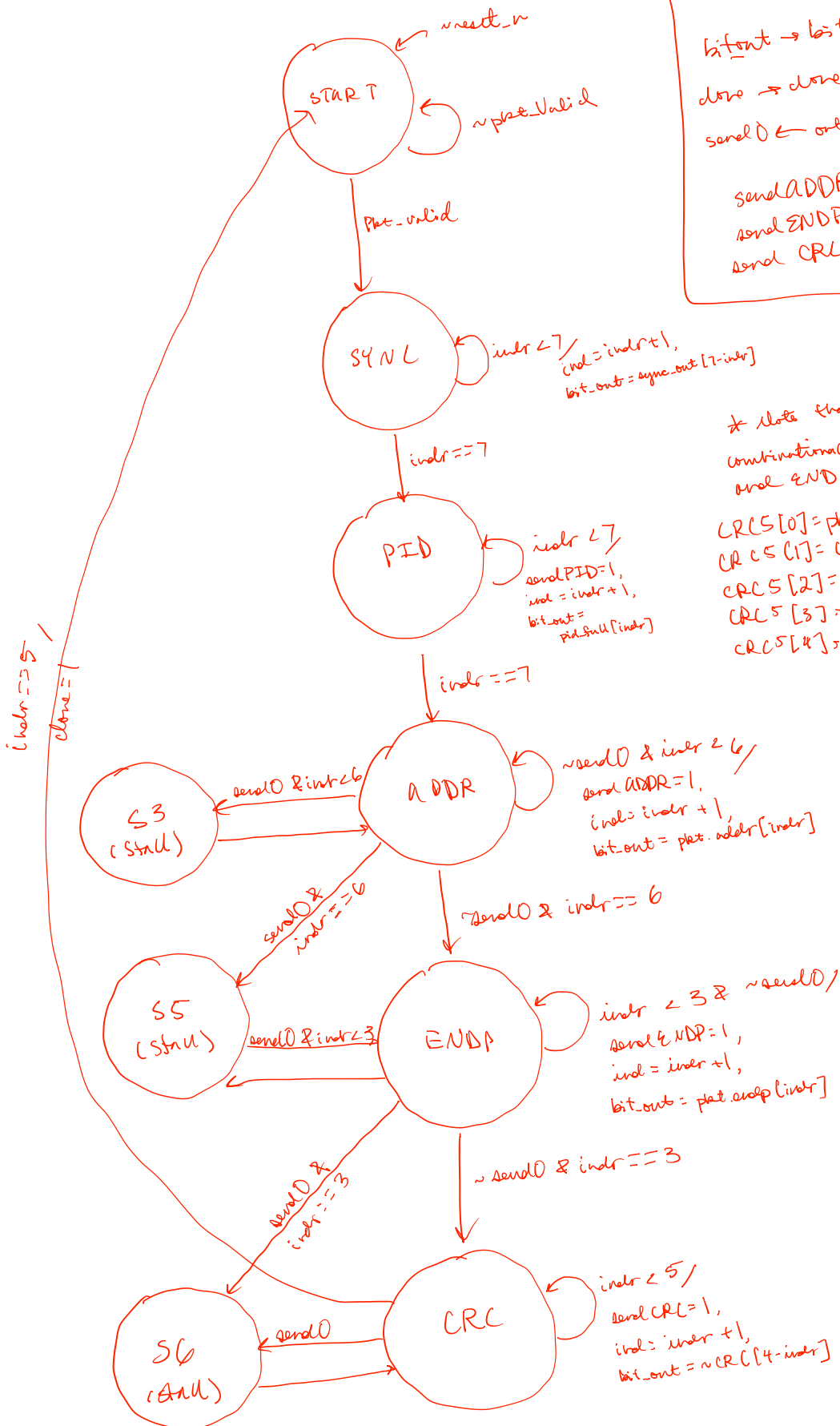
$$CRC5[0] = \text{plat_addr}[\text{indr}] \oplus CRC_r[4]$$

$$CRC5[1] = CRC_r[0]$$

$$CRC5[2] = (\text{plat_addr}[\text{indr}] \oplus CRC_r[4]) \oplus CRC_r[1]$$

$$CRC5[3] = CRC_r[2]$$

$$CRC5[4] = CRC_r[3]$$



BS FSM



handshake signals:

bit_out $\rightarrow$ bit_in input for NRZ1
 bit_in $\leftarrow$ bit_out output from CRC
 send0 $\rightarrow$ send0 input for CRC
 sendADDR $\leftarrow$ sendADDR output from CRC
 sendENDP $\leftarrow$ sendENDP output from CRC
 sendCRC $\leftarrow$ sendCRC output from CRC

bit_in != 1 /
 send0 = 0

* if we receive consecutive 1s, send out a 0

NRZI FSM

handshake signals:

done $\leftarrow$ done output from CRC
bit.in $\leftarrow$ bit-out output from BS

